

Highly-Efficient Recycling of Poly(ethylene terephthalate)/Polyethylene Film through Low-Temperature Methanolysis Catalyzed by Nonmetallic Deep Eutectic Solvent

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While the poly(ethylene terephthalate) (PET)/polyethylene (PE) multilayer films find extensive applications, particularly in packaging, the recovery of single PE layer or high-purity monomers from these films is seriously hindered by their complex compositions. Herein, a novel low-temperature methanolysis strategy for the highly efficient recycling PET/PE multilayer films with nonmetallic deep eutectic solvent (DES) is developed, where PET can be depolymerized to dimethyl terephthalate (DMT) and ethylene glycol (EG) monomers at 115 °C and 0.45 MPa for 60 min with 100% PET conversion and up to 97.1% DMT yield while PE maintains its stability. The process flow for the high-efficient recycling of

PET/PE multilayer films is proposed, consisting of the methanolysis step and two solid–liquid separation ones to obtain the PE, DMT, and EG products. The process optimization, catalyzed mechanism, and swelling behaviors of DES for the PET methanolysis are studied. Finally, the kilogram-scale experiments under best conditions also can obtain 100% PET conversion and 87.8% DMT yield with complete recovery of PE, which efficiently confirms scalability of the proposed recycling pathway. This work establishes a sustainable pathway for closed-loop recycling of multilayer plastics by integrating PE material recovery with PET chemical upcycling.

1. Introduction

The exponential growth of global plastic production, surging from 2 million tons in the 1950s to over 500 million tons in 2025, has precipitated the serious environmental crisis driven by inadequate disposal and low recycling rates.^[1–3] Among various plastics, multilayer plastic films, engineered through the lamination of distinct polymer types into a single composite structure, find extensive applications, particularly in packaging field. These films offer superior functionality compared to single-material alternatives by integrating the specific performance characteristics of their constituent polymers, such as barrier properties, mechanical strength, and thermal stability. The global multilayer films reach USD 61.3 billion by 2025, representing a compound annual growth rate of 4.3% from 2020.^[4] Despite these functional advantages, concerns regarding the environmental impact of post-consumer multilayer film waste are escalating, necessitating effective recycling solutions.^[5] PET/PE films dominate the multilayer film market, accounting for >50% of global usage, due to their synergistic material properties, that is, PET provides exceptional aroma retention and thermal stability (resisting both heat and cold), while PE contributes crucial impact resistance. Therefore, the recycling of

PET/PE films is critical for mitigating plastic waste impacts and advancing global sustainable packaging goals.

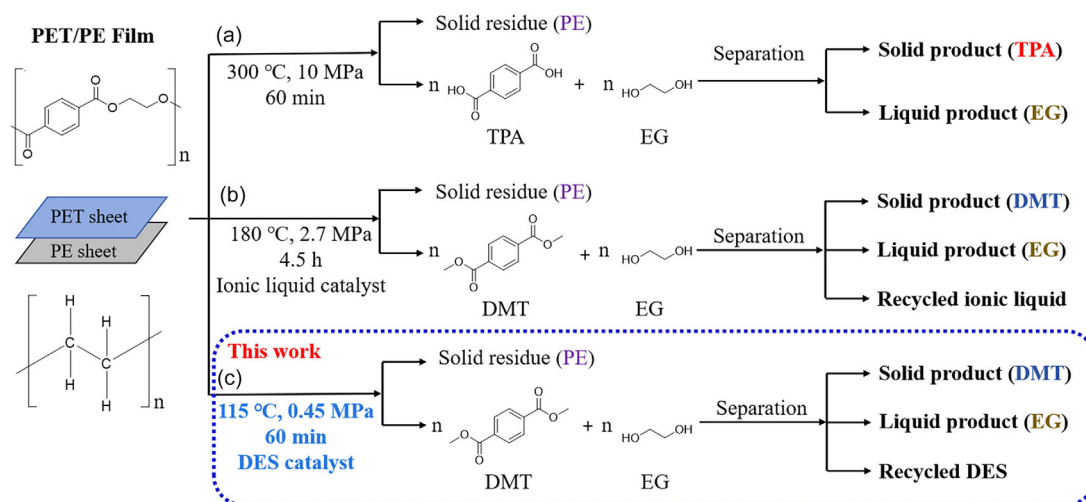
At present, PET/PE film recycling mainly includes mechanical and chemical methods. However, conventional mechanical recycling methods of PET/PE films, such as melt extrusion, reformation, and pelletization, face fundamental limitations, including color darkening and deterioration of the mechanical properties of the recycled plastics. For instance, the inherent immiscibility of PET and PE results in recycled products with heterogeneous blends, leading to phase separation and degraded material properties that restrict practical applications. Chemical recycling has emerged as a transformative and promising alternative, depolymerizing PET/PE films into monomers like terephthalic acid (TPA), dimethyl terephthalate (DMT), or ethylene glycol (EG) while keeping PE stable through hydrolysis, glycolysis, or methanolysis.^[6,7] The hydrothermal treatment of PET/PE films can selective PET degradation to TPA and EG while preserving PE solid stably which can be separated from the reactor, but it demands high temperature and pressure (300 °C and 10 MPa) (Scheme 1a).^[8] The PET glycolysis employing EG under mild conditions remains commercially prevalent, but still requires high temperatures (>190 °C), prolonged reaction times, and metal ion contamination (e.g., zinc acetate), larger formation of oligomers, and even more serious Bis(2-hydroxyethyl) terephthalate purification challenge.^[9–11]

In this context, the low-temperature and high-efficient methanolysis of PET with nonmetallic catalysts, such as ionic liquids (ILs)^[12,13] and deep eutectic solvents (DESs),^[14,15] has emerged as a promising avenue to mitigate these limitations, due to the easy purification of DMT monomer. ILs like BMIMOH enable selective PET methanolysis to DMT and EG at 180 °C while preserving PE, allowing simultaneous separation and recycling of mixed plastics

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Scheme 1. Methods of recycling PET/PE film. a) Hydrolysis. b) Methanolysis. c) Methanolysis.

(Scheme 1b).^[16] However, the methanolysis of PET catalyzed by ILs still employs harsh reaction conditions like high reaction temperature and high reaction pressure. Alternatively, DESs, which are composed of hydrogen bond acceptors (HBAs) and donors (HBD), have gained traction as low-cost, sustainable, and high-efficient catalysts for the PET recycling. DES-based PET glycolysis/methanolysis with minimal energy consumption, yielding high-purity monomers and avoiding toxic solvents. For example, Tang et al. synthesized a choline chloride (ChCl)/Zn(OAc)₂ DES catalyst and introduced acetonitrile as a co-solvent for PET methanolysis. This approach enabled complete PET degradation at the significantly milder temperature of 120 °C within 2 h, although the DMT yield was limited to 80%.^[17] In addition, Li et al. developed a 1,5-diazabicyclo [4.3.0]-5-nonene (DBN)/Phenol catalyst for PET methanolysis with 100% PET conversion and 95.3% DMT yield at 130 °C for 1 h,^[14] which proves the high-efficient catalysis of DES for PET methanolysis. However, to the best of our knowledge, there is currently no report regarding to the recovery of PET/PE multifilms using PET methanolysis with metal-free DES catalyst.

From this sense, the novel low-temperature methanolysis strategy with high-efficient nonmetallic DBN/5-methyl-2-Isopropylphenol (blf) DES system was developed with the methanolysis step and two solid–liquid separation ones for separation of PET/PE films, achieving 100% PET conversion at 115 °C surpassing traditional metal-free systems, as displayed in Scheme 1c. Under specific reaction conditions, PET is completely methanolized by DBN/blf catalyst into DMT and EG monomers which are well dissolved in high-temperature methanol/DES system. The unreacted solid PE can be easily, which is the first solid–liquid separation step under methanolysis condition. After the resultant liquid phase cooling, DMT precipitates due to its low solubility in methanol and can be separated from liquid products containing EG, which is referred to as the second solid–liquid separation step under room condition. The methanolysis process of PET/PE multilayer films with two solid–liquid separation steps using methanol as the only solvent is expected to recover PE and PET monomers separately, thereby achieving simultaneous material and chemical

recycling of PET/PE multilayer films. This work not only addresses the critical challenge of PET/PE film recycling but also aligns with global sustainability goals by reducing energy consumption and eliminating hazardous solvents.

2. Results and Discussion

2.1. The Process Optimization of PET Methanolysis

The effects of three DBN-based DESs (DBN/blf, DBN/EG, and DBN/Oac) as catalysts on PET methanolysis were initially explored (Figure 1a). DBN/blf exhibits the best catalytic activity with the highest yield (45.5%) of DMT compared with DBN/EG, and DBN/Oac at the same conditions (115 °C, 60 min, *m*MeOH:*m*PET/PE = 8, *m*DES:*m*PET/PE = 0.05), superior to metal salt catalysts (Zn(OAc)₂, K₂CO₃, and KOAc). Specially, compared with these metal salt catalysts, the reaction process will not produce byproducts containing heavy metals. The reason for better catalytic performance of DBN/blf is probably attributed to its smaller the highest occupied molecular orbital-lowest unoccupied molecular orbital (HOMO-LUMO) gap and the situation that the OH-group in blf is more likely to form hydrogen bonds with the nitrogen (N) atom on DBN, thereby increasing the electronegativity of the N atom (Figure 1b).^[18] Furthermore, the DMT yield catalyzed by DBN/blf is larger than that catalyzed by other DBN-based DESs, which indicates that the presence of co-catalyst components and the formation of DBN/blf facilitate the stronger capture of hydroxyl hydrogen from MeOH by HBAs.^[19]

The influence of different reaction factors on PET methanolysis under DBN/blf catalyst is further discussed in Figure 1c–e, including temperature, methanol amount, and catalyst amounts. With the increased temperature, the conversion rate of PET and the yield of DMT increase significantly. However, during the heating process, the r-PE (recycled PE) fragments undergo swelling and mutual adhesion which cause difficulty in separating PE from the reactor at 125 °C, while the r-PE fragments show slight change

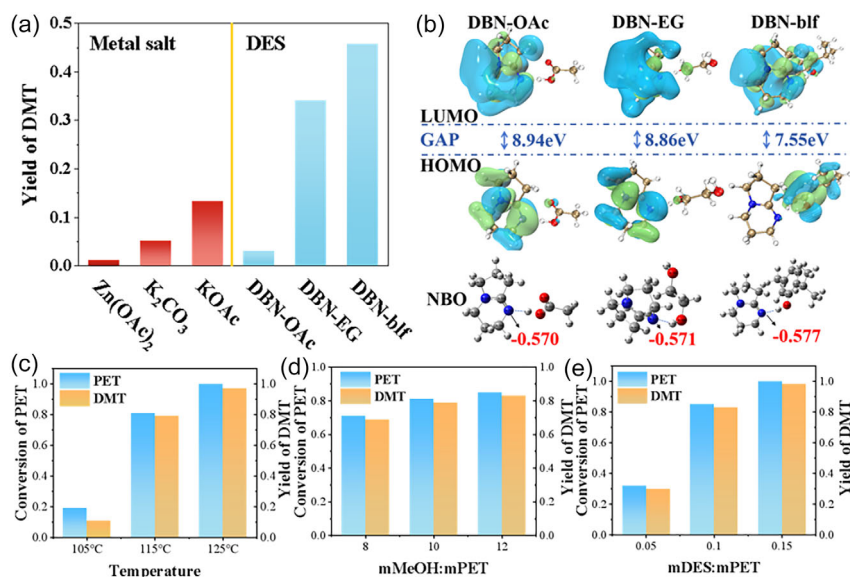


Figure 1. a) The yield of DMT under Zn(OAc)₂, K₂CO₃, KOAc, DBN/OAc, DBN/EG, and DBN/blf at 115 °C, 1 h, mMeOH:mPET/PE = 8, mDES:mPET/PE = 0.05. b) HOMO, LUMO, and HOMO-LUMO gap of DBN/OAc, DBN/EG, and DBN/blf, and the NBO charge density of N atom in DBN. C atoms (dark gray), O atoms (red), H atoms (light gray), and N atoms (blue), wherein the dashed part represents the hydrogen bond formed by the ion pair. c-e) The results of PET conversion, DMT yield at different temperatures, mMeOH:mPET/PE, mDES:mPET/PE.

under reaction condition of 115 °C (Figure S3, Supporting Information), followed by consolidation into a monolithic solid upon cooling-induced contraction. When the catalyst dosage is low (5%), the PET conversion rate and DMT yield are only 32.2% and 30.3%, respectively. Increasing the catalyst dosage up to 15 wt%, the DMT yield reaches 97.1% with 100% PET conversion rate. At the mass ratio of methanol to composite film of 8:1, the PET conversion rate and DMT yield are 71.8% and 68.7%, respectively. When the mass ratio of methanol dosage to composite film was increased to 1:12, both PET conversion (85.6%) and DMT yield (82.5%) show slight improvements. Thus, combined with the results of orthogonal experiments (Table S1, Supporting Information), the yield of DMT can be up to 97.1% with 100% conversion of PET (115 °C, 60 min, mMeOH:mPET/PE = 12, mDES:mPET/PE = 0.15).

2.2. Methanolysis Mechanism of PET

The specific mechanism is illustrated in Figure 2a. DBN/blf interacts with methanol, generating methoxide anion structures. These enhance the electrophilicity of the carbonyl carbon of PET. Subsequently, the methoxide anion initiates a nucleophilic attack on the carbonyl carbon of PET. This leads to the cleavage of the C—O bond in PET, simultaneously forming DMT and EG.

To prove the reaction mechanism of PET methanolysis under DBN-blf, the in situ-Fourier transform infrared (FTIR) under optimal reaction condition is shown in Figure 2b. The peaks at 3586, 1050, 1030, and 1019 cm⁻¹ correspond to the O—H stretching vibration of methanol, the C—H bending vibration of terminally adsorbed methanol, the C—H bending vibration of doubly-bridged adsorbed methanol under van der Waals forces, and the C—H bending vibration in methoxide anions, respectively. As the

reaction proceeds, the O—H bond of methanol cleaves, and its characteristic peak gradually disappears, while the structure of the generated methoxide anions becomes increasingly evident. The peaks at 1154, 1302 (1338), and 1742 cm⁻¹ correspond to the C—O stretching, C—H out-of-plane (wagging) bending, and C=O stretching vibration in PET, respectively. The peaks at 1213, 1316 (1365), 1764, and 2954 cm⁻¹ correspond to the C—O stretching, C—H out-of-plane (wagging) bending, C=O stretching, and C—H stretching vibration in DMT, respectively. As the reaction progresses, PET gradually converts to DMT, indicated by the replacement of the C—O signals and the increasingly distinct C—H stretching vibration assigned to DMT. Finally, the broadening in the region of 3426 to 3460 cm⁻¹ confirms the formation of O—H groups in EG.

As correlated with the predicted HOMO-LUMO energy gaps in Figure 1b, the peaks at 1678 and 1646 cm⁻¹ in Figure 2c are attributed to the C=N stretching vibration in DBN (peak splitting occurs due to the coupling of C=N and C—N vibrations). After the formation of the DES between DBN and blf, the split peak at 1678 cm⁻¹ disappears. This indicates that the nitrogen atom in the C=N structure of DBN forms the hydrogen bonds with blf, leading to the red-shift and merging of the split peak at 1678 cm⁻¹. Corresponding to the predicted HOMO-LUMO molecular energy gap in Figure 1b, in DBN-EG, the O—H peak at 3341 cm⁻¹ is partially retained. This is due to insufficient proton transfer during hydrogen-bonding formation between EG and the DES components, resulting in a higher GAP energy (8.86 eV), which consequently leads to suboptimal catalytic performance of this DES. In DBN-OAc, the split peak at 1645 cm⁻¹ disappears, and the intensity of the split peak at 1678 cm⁻¹ increases. This is attributed to the lower bond energy of the O—H bond in OAc, making the system prone to generate the HBA CH₃COO⁻, which competes with DBN, leading to instability in the DES structure.

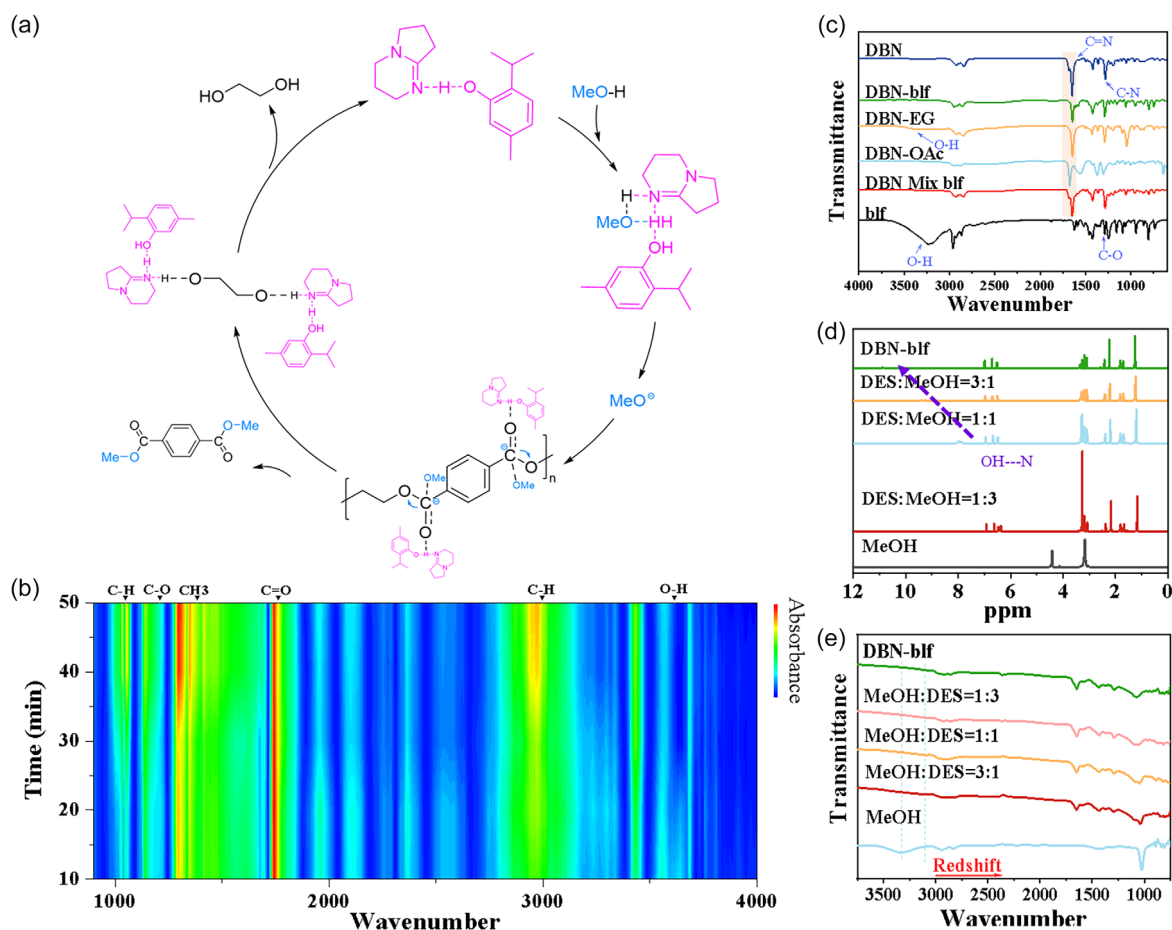


Figure 2. a) Specific mechanism of PET methanolysis under DBN-blif catalyst. b) The results of in situ-FTIR spectra. c) FTIR spectra of DBN, DBN-blif, DBN-EG, DBN-OAc, DBN Mix blf, and blf. d) ^1H NMR spectra. e) FTIR spectra for DBN/blif, MeOH, and their mixtures.

The interactions between the hydroxyl hydrogen of MeOH and DBN/blif were investigated by ^1H NMR and FTIR.^[20] A gradual movement of the hydroxyl hydrogen signals for MeOH toward the downfield, accompanied by a broadening and eventual near disappearance of the peak with increasing concentration of DBN/blif (Figure 2d). Specifically, in the mixed solutions of MeOH and DBN/blif, the hydroxyl hydrogen signal peak undergoes a shift from 7.98 to 10.89 ppm as the molar ratio of DBN/blif to MeOH increases from pure DBN/blif, 3:1 to 1:3 and pure MeOH. As the molar ratio of DBN/blif versus MeOH increases from pure DBN/blif, 3:1 to 1:3 and pure MeOH, the $-\text{OH}$ vibration of MeOH gradually broadens and nearly disappears in the FTIR spectrum (Figure 2e). The redshift of $\text{N}-\text{H}$ in DES reaches 105.0 cm^{-1} , from 3317.1 to 3212.1 cm^{-1} . These all were ascribed to the formation of robust hydrogen bonds between the nitrogen atom in DBN and the hydrogen atom of the hydroxyl group in MeOH, consistent with reported results.^[21–24]

2.3. The Process Flow for PET/PE Film Separation

The proposed process flow for the separation of PET/PE film through low-temperature methanolysis is displayed (Figure 3a). Methanol, catalyst, and feedstock are fed into a reactor, and then

the reaction mixture undergoes primary methanolysis and subsequent solid–liquid separation to remove PE (polyester oligomers/impurities) to obtain the DMT-containing filtrate. This filtrate is subsequently cooled, followed by the secondary solid–liquid separation to isolate the solid DMT product from the solvent. The solvent stream is then directed to a distillation unit for primary distillation. Here, the methanol is separated from high-boiling-point liquids (EG and DES), which are recycled back to the reactor. Finally, the high-boiling-point liquid fraction undergoes vacuum distillation to separate EG from DES catalyst. The DES is returned to the process for reuse. After all these procedures, “solid residue” (white color), “solid product” (white color), and “liquid product” (transparent solution) were collected (Figure 3b). More details about the raw materials, apparatus system, experimental procedure, and analysis methods are described in Supporting Information.

Figure 3c,d shows the identification results of the solid residue, solid product, and liquid product obtained from the recycling of PET/PE films at 115°C and 0.45 MPa for 60 min. From Figure 2c, the FTIR spectrum of the solid residue shows peaks at 2910 and 2850 cm^{-1} due to the characteristic $\text{C}-\text{H}$ stretching vibration of PE and peaks at 1460 and 720 cm^{-1} due to the $\text{C}-\text{H}$ bending vibration of PE. Comparing the standard peaks of pure

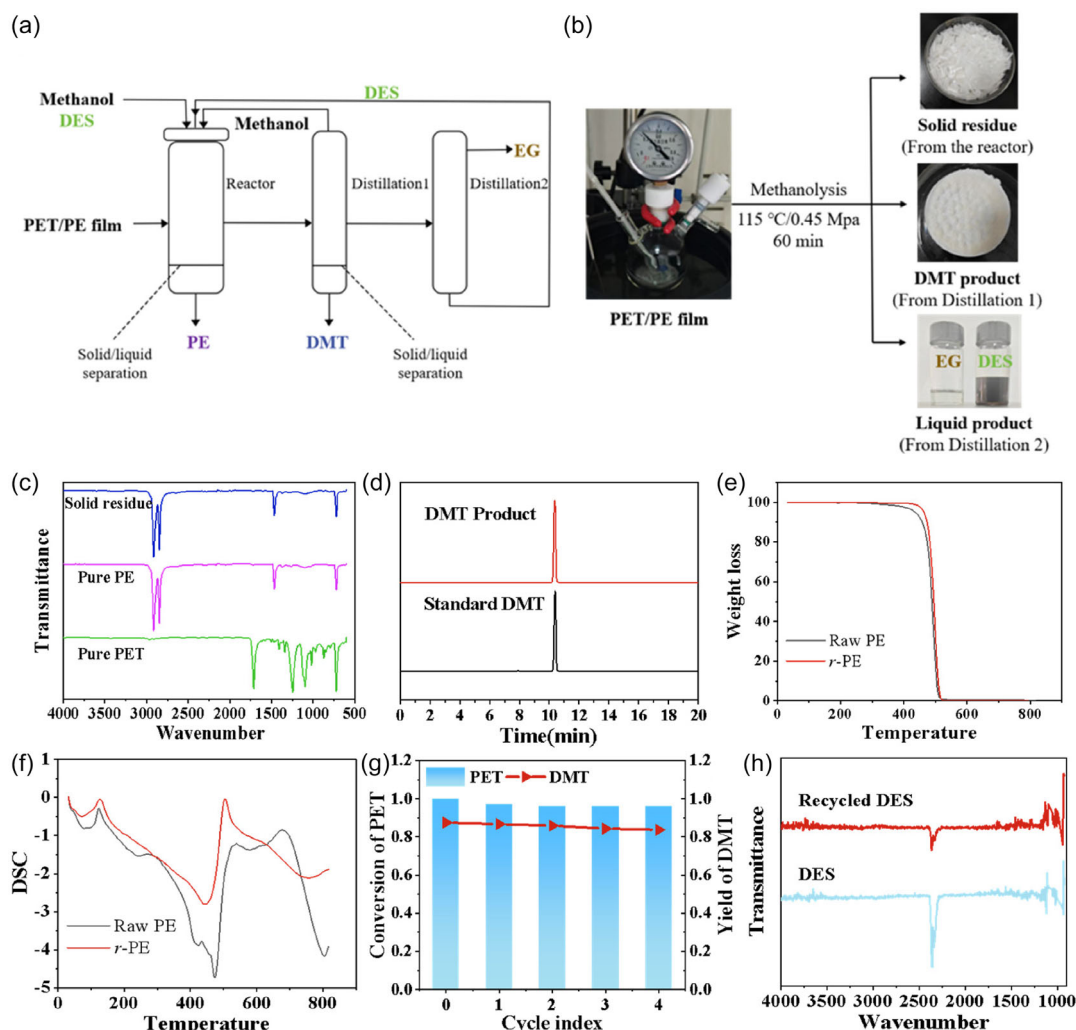


Figure 3. a) Schematic diagrams of the process flow for the separation of PET/PE film. b) the obtained PE, DMT, and EG products. c,d) FTIR spectra and GC spectra of the PE and DMT solid product. e,f) DSC and TG spectra of raw PE and PE after reaction under different temperature. g) The results of PET conversion, DMT yield in 4-cycle experiment. h) FTIR spectra of DES and recycled DES.

PE and PET, the peaks of the solid residue were closely consistent with PE, but entirely distinct from PET, which suggests that the solid residue consists primarily of PE in PET/PE films. The solid product DMT shows the same peak position of the gas chromatography (GC) and FTIR spectra as the standard DMT (Figure S4, Supporting Information), in which, the peaks at 1670, 1510, 1420, 1280, and 920 cm^{-1} are attributed to the characteristic C=O stretching of the carboxyl group, the C=C stretching of the aromatic ring, the C—O—H out-of-plane bending, and the C—O stretching, respectively. All peaks of the solid product correspond to those of pure DMT.

There were slight differences in the thermodynamic properties between *r*-PE and Raw-PE (Figure 3e). Raw-PE starts to lose weight at about 240 °C during the heating process from 20 °C to 800 °C, and reaches the maximum weight loss rate at 510 °C, finally retaining 0.82% of its mass. In addition, there is a significant mass decrease starting at 301 °C, and up to 450 °C, the mass of PE begins to decrease sharply, reaching the maximum weight

loss rate at 510 °C, and finally retaining 0.71% of its mass. Both of the raw-PE and *r*-PE exhibit two distinct endothermic peaks (Figure 3f). The first endothermic peak of *r*-PE at 167 °C coincides with that of raw-PE and the second peak *r*-PE at 459.8 °C displays a marginal rightward shift in relation to that of Raw-PE at 455.1 °C. These findings collectively indicate that *r*-PE retains essentially identical properties to their respective raw-PE counterparts under 115 °C, indicating that no significant alterations in the structural and property characteristics of PE occur throughout the methanolysis procedure. Furthermore, the catalytic activity of DBN/bif catalyst exhibits no significant decline after four cycles, with the DMT yield decreasing by only 3.8% (Figure 3g and Figure S5, Supporting Information).

In addition, the FTIR characteristic peak positions of the catalyst recovered via vacuum distillation are identical to those of the original catalyst (Figure 3h), indicating that the recycled catalyst retained its structural integrity. The only slight drawback is that according to Figure S6, Supporting Information, the peak at

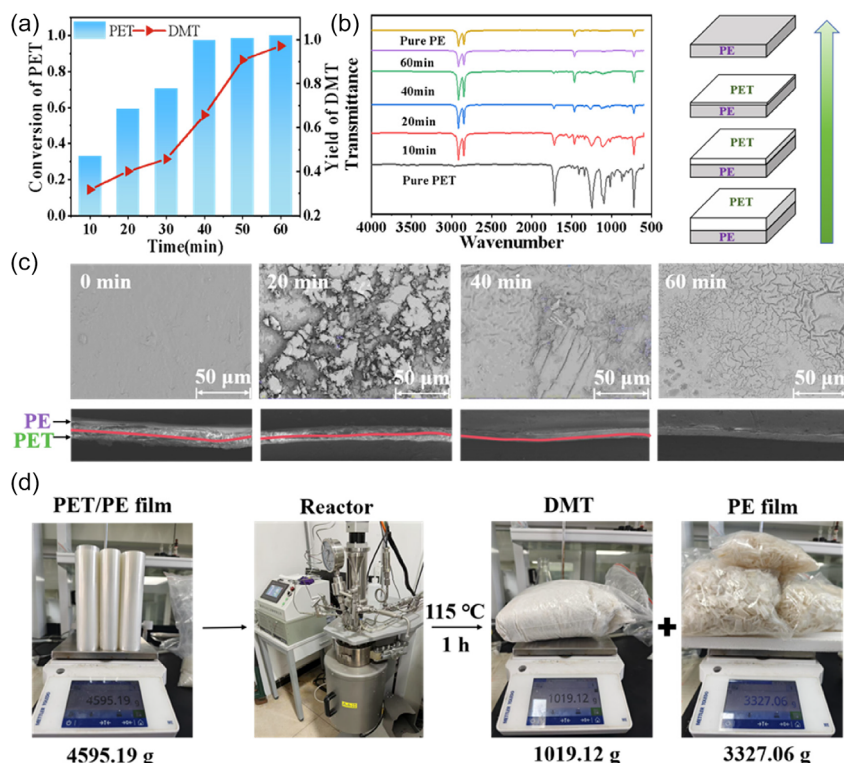


Figure 4. a) The results of PET conversion, DMT yield at different reaction time. b,c) FTIR spectra and SEM images of solid residues obtained at different holding times. d) Kilogram-scale experiment.

63.5 ppm in the ¹³C NMR analysis corresponds to the carbon atom of EG. This indicates that EG, a product of the PET methanolysis reaction, becomes incorporated into the recycled DBN-blif catalyst, diluting the original DES and causing the decline in catalytic performance of the recycled DES.

2.4. Morphological Characteristics and Kilogram-Scale Experiments

For a more accurate evaluation of the process, the time-dependent amount of decomposed PET was calculated, and the corresponding amounts of DMT were quantified (Figure 4a). Under the optimal reaction conditions, PET conversion increases from 0 to 10.9% in the first 10 min and then rises to 90.4% at 40 min. After 40 min, PET conversion increases slowly with extended reaction time, and finally reaches 100.0% at 60 min. For the FTIR spectra of the solid residues obtained from 0 to 60 min (Figure 4b), the peaks of PET other than PE become weaker and finally disappear as the reaction time increases. These peaks are identified to be consistent with the characteristic peaks of pure PET, such as the ester C=O stretching peak at 1710 cm⁻¹, the C–O stretching peak at 1240 cm⁻¹, and the C–H bending peak at 1020 cm⁻¹, which indicates that ET in PET/PE film decomposes with reaction time.

To systematically investigate the effect of DBN/blif catalyst on PET/PE depolymerization process, the morphological characteristics of the residual solid over time during PET methanolysis were further investigated by scanning electron microscope (SEM). From

Figure 4c, in the presence of the catalyst, a few cracks appear on the surface of PET (20 min), and corrosion pits are formed. When reaction time is prolonged to 40 min, the PET is further mutilated with large amounts of etching holes on the surface, and after 60 min reaction, PET basically disappears. Meanwhile, the cross-section shows the thicknesses of PET film are reduced from 32.16 μm (original PET/PE film) to 15.13 μm (60 min). These findings indicate the well-swelling behaviors of DES on PET surface. Finally, kilogram-scale experiments were conducted in a 3 L reactor, achieving 100% PET conversion and 87.8% DMT yield (Figure 4d), which further validates the industrial viability of DES-catalyzed methanolysis for PET/PE composite films, providing critical insights for advancing the industrial-scale recycling of such multilayer materials.

3. Conclusion

In this work, an efficient and sustainable pathway for recycling PET/PE multilayer films using methanolysis process catalyzed by a nonmetallic DES (DBN/blif) was proposed. Operating under mild conditions (115 °C, 0.45 MPa, 60 min), the system achieves complete PET conversion (100%) and near-quantitative DMT yields (97.1%). The DBN/blif DES exhibits exceptional catalytic activity due to its narrowed HOMO-LUMO gap and enhanced electronegativity of nitrogen atoms, facilitating robust hydrogen bonds with methanol to activate the PET methanolysis. Critically, the process

enabled simultaneous recovery of high-purity PE and PET monomers through a novel two-step solid–liquid separation: (1) selective dissolution of PET-derived DMT/EG in hot methanol while preserving PE as a solid residue, and (2) crystallization of DMT upon cooling. Recovered PE (*r*-PE) retains structural integrity and thermal properties, confirming minimal degradation. Kilogram-scale validation achieves 100% PET conversion with 87.8% DMT yield, underscoring scalability. This approach overcomes key limitations of existing PET/PE recycling methods, which provides a closed-loop solution for multilayer plastic waste—concurrently enabling material recycling of PE and chemical upcycling of PET into high-value DMT.

4. Experimental Section

Materials

The PET/PE sample was provided by Cangzhou Zhongxin Plastic Industry Co., Ltd, which is a laminated film of 25% PET and 75% linear low-density PE with a total thickness of 32 μm . Methanol ($\geq 99.9\%$), acetic acid (Oac, 99%), and 4-Methylimidazole (99%) were obtained from Shanghai Macklin Biochemical Co. Dimethyl sulfoxide- d_6 (DMSO- d_6 , 99.5%) and 1,5-diazabicyclo [4.3.0]non-5-ene (DBN, $\geq 98\%$) were obtained from Shanghai Titan Scientific Co.

Synthesis of DBN-Based DESs

The various DBN-based DESs were synthesized. DBN and blf were mixed in a ratio of 1:1, and then, the reaction products were heated to 70 $^{\circ}\text{C}$ for 6 h with stirring. After completion, the reaction solution was cooled to room temperature, and the water was removed by drying it in a vacuum oven at 60 $^{\circ}\text{C}$ for 24 h. Finally, the target DES was obtained. DBN/EG, DBN/Oac were prepared according to the above method.

Instrumentation

Nuclear magnetic resonance measurements were performed on Bruker Advance instrument at 600 MHz (^1H NMR, ^{13}C NMR) for DES catalysts and methanol solutions. Online infrared monitoring system (Online FTIR, Mettler Toledo React IR) was used to monitor the model compound in the reaction system over a wavenumber range from 600 to 4000 cm^{-1} . Thermal gravimetric analysis (TGA) and differential scanning calorimetry (DSC) measurements were performed on a TG-DSC Instrument (Netzsch STA 409PC, Germany) under nitrogen atmosphere. PE samples were heated from ambient temperatures to 900 $^{\circ}\text{C}$ at a heating rate of 10 $^{\circ}\text{C min}^{-1}$. The morphological characteristics of PET films during the depolymerization process were observed with SEM (Hitachi, Regulus8100). GC measurement was performed at Agilent 7890B Gas Chromatograph. Prior to analysis, samples were dissolved in acetonitrile. The sample temperature was 250 $^{\circ}\text{C}$, and the column temperature was 35 $^{\circ}\text{C}$. The temperature program was as follows: hold at 35 $^{\circ}\text{C}$ for 7 min, then ramp to 300 $^{\circ}\text{C}$ at a rate of 15 $^{\circ}\text{C min}^{-1}$. Hold at 300 $^{\circ}\text{C}$ for 10 min, then cool down to ambient temperature.

PET Depolymerization Process

Dry PET/PE composite film is first shredded into thin strips ≈ 1 cm wide as a pre-treatment. A measured quantity of this shredded film is then combined with methanol and DES in a pressure-resistant flask

equipped with a magnetic stir bar, and the mixture is stirred at ≈ 400 rpm. Upon completion of the reaction, the mixture is filtered while hot. The recovered unreacted solids are subsequently soaked in acetonitrile for 12 h to dissolve any residual DMT adhering to them, followed by a second filtration. These unreacted solids are then dried under vacuum at 80 $^{\circ}\text{C}$ for 12 h and weighed. The conversion of PET is defined as

$$\text{Conversion of PET} = \frac{4(W_0 - W_1)}{W_0} \times 100\% \quad (1)$$

where W_0 is the initial mass of PET/PE film, and W_1 is the mass of the unreacted solid. The initial filtrate, containing DMT, methanol, EG, and DES, is concentrated by rotary evaporation. The resulting residue is washed with a methanol/water solution (1:1 mass ratio) and filtered. The obtained solid (DMT) is dried at 80 $^{\circ}\text{C}$ for 12 h and weighed. The calculation formula is as follows

$$\text{Yield of DMT} = \frac{W_{\text{DMT}}/M_{\text{DMT}}}{W_{\text{PET}}/M_{\text{PET}}} \times 100\% \quad (2)$$

where W_{DMT} represents the weight of the main product after the reaction. Conversely, M_{PET} and M_{DMT} represent the relative molecular weight of the PET repeating unit (192 g mol^{-1}) and the molecular weight of DMT (194 g mol^{-1}), respectively. The filtrate from this washing step is concentrated by rotary evaporation to yield a mixture of EG and DES. EG is finally removed from this mixture via reduced-pressure distillation at 170 $^{\circ}\text{C}$ and -0.1 MPa, resulting in the recovery of the DES.

DFT Calculation

All density functional theory (DFT) calculations were carried out using the M06-2X^[25] functional in conjunction with the 6-311+G(d,p) basis set by Gaussian 09 programs.^[26] The def2-TZVP^[27] basis set was used for single-point energy calculation. To confirm the transition state structures, frequency calculations were carried out at the same level of theory to ensure only one imaginary frequency. To include the solvent effects, the solvation model based on density had been performed with the dielectric constant of MeOH ($\epsilon = 32.7$). To eliminate the dispersion effect, the empirical dispersion keyword = gd3 was added to the route section. Multiwfn code^[28] was used to analyze the HOMO, LUMO, and natural bond orbital (NBO).

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Conflict of Interest

The authors declare no conflict of interest.

Author Contributions

Xiashi Wang: data curation (lead); formal analysis (lead); investigation (lead); validation (lead); writing—original draft (lead). **Xin Wei:** methodology (equal); software (equal); visualization (equal); writing—original draft (equal). **Jingwen Qiu:** investigation (supporting); methodology (supporting). **Weizhong Zheng:** resources

(equal); software (equal); writing—review and editing (lead). **Weizhen Sun**: software (equal); supervision (equal). **Ling Zhao**: resources (equal); software (equal); supervision (equal). **Xiashi Wang** and **Xin Wei** contributed equally to this work.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Keywords: deep eutectic solvent • methanolysis • Poly(ethyleneterephthalate)/Polyethylene multilayer films

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